

Airline Passenger Satisfaction Analysis

Objective: identify factors driving satisfaction, loyalty, and travel experience to inform actionable recommendations.





Project Snapshot

Dataset

130K+ passengers: demographics,
travel info, service ratings,
delays, satisfaction.

Tools

Power BI · DAX · Power Query ·
Data Modeling · Visualization

Goal

Reveal drivers of satisfaction and recommend improvements.

Executive Summary — Key KPIs

130K

Total Passengers

71K

Satisfied Passengers

54.73%

Satisfaction Rate

2.84/5

Seat Comfort



Customer Segments



Loyalty & Travel Type

- Loyal customers $\approx$ 82% of passengers
- Business travel $\approx$ 69% of trips
- Business travelers show higher satisfaction

❏ Recommendation: prioritize loyalty programs and personalized engagement for high-value travelers.

Class Differences



Business Class passengers report higher satisfaction than Economy. Experience gap driven by comfort, amenities, and service levels.



Recommendation: improve Economy Class service standards to reduce the satisfaction gap.

Service Performance — Strengths



Cleanliness

Average rating: 3.71 —
consistent positive feedback.



Baggage Handling

Average rating: 3.70 —
dependable performance
appreciated by passengers.



Online Support

Average rating: 3.52 —
exceeds expectations in
digital channels.

Service Performance — Improvement Areas



Seat Comfort

Lowest-rated service: 2.84/5
— top priority for upgrades.



Food & Drink

Average rating: 2.85 — menu
quality and variety need
improvement.



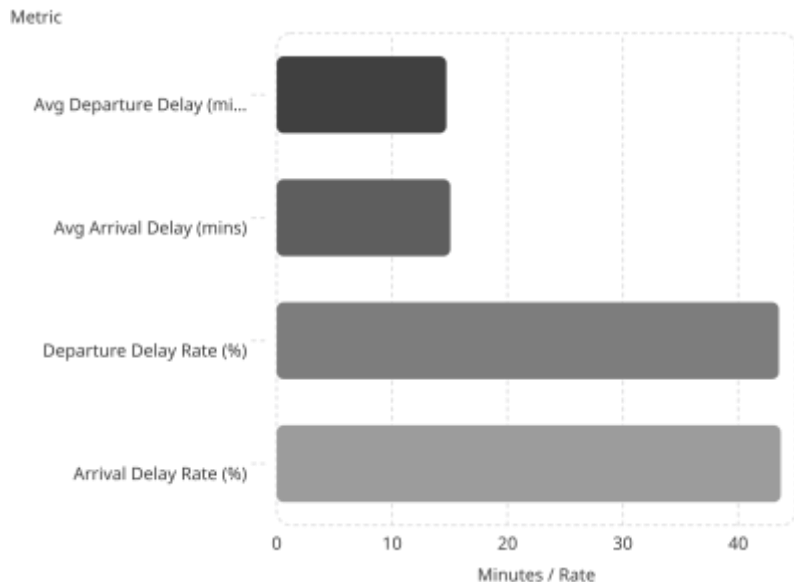
Gate Location

Average rating: 2.99 —
airport wayfinding and
accessibility issues.



Action: prioritize seat and food upgrades plus better gate signage.

Operational Insights — Delays

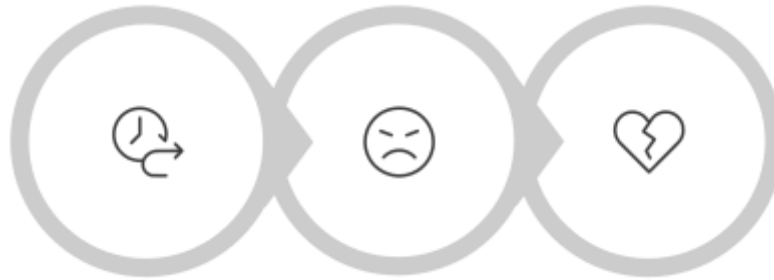


Delays correlate with lower satisfaction. Longer delays → significantly reduced scores.



Recommendation: optimize scheduling and turnaround to reduce delay impact.

Additional Insights & Target Groups



Delay Rises

Satisfaction Falls

Loyalty Weakens

Delays erode satisfaction and threaten loyalty over time.

Distance & Age Insights

- Long-haul passengers show higher dissatisfaction due to longer travel expectations.
- Age 31-45 is the largest segment and key contributor to satisfaction metrics.

✓ Recommendation: enhance long-haul comfort and target 31-45 with personalized offers.



Strategic Recommendations & Conclusion

Short-Term

Improve seat comfort, upgrade food quality, enhance gate navigation.

Medium-Term

Reduce flight delays, strengthen support channels, raise Economy standards.

Long-Term

Invest in cabin upgrades, expand loyalty programs, implement predictive delay analytics.

Conclusion: Focus on service quality (cleanliness, baggage, support), address seat comfort and delays to improve satisfaction, loyalty, and passenger experience.